Math 257 Definitions and Theorems

Week 10: Modules 33-34 and Linear Differential Equations

Fall 2023

33 Powers of Matrices

Key idea: If a matrix A has an eigenbasis, then it is easy to compute A^k, even for large k.

Theorem 33.1. *If $A = PDP^{-1}$, where D is a diagonal matrix, then for any m,*

$$A^m = PD^mP^{-1}.$$

34 Matrix Exponential

Definition 34.1. Let A be an $n \times n$-matrix. We define the **matrix exponential** e^{At} as

$$e^{At} = \sum_{k \geq 0} \frac{(At)^k}{k!} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \dots$$

Theorem 34.1. *Let A be an $n \times n$ matrix. Then*

- *the series e^{At} always converges,*
- $e^{At}e^{-At} = I_n$,
- $e^{At}e^{As} = e^{A(t+s)}$,
- $\frac{d}{dt}(e^{At}) = Ae^{At}$.

Theorem 34.2. *Let A be a $n \times n$ matrix such that $A = PDP^{-1}$ for some invertible matrix P and some diagonal matrix D. Then*

$$e^{At} = Pe^{Dt}P^{-1}.$$

Linear Differential Equations

Definition 34.2. A **linear (first order) differential equation** is an equation of the form

$$\frac{d\mathbf{u}}{dt} = A\mathbf{u}$$

where $\mathbf{u}$ is function from $\mathbb{R}$ to $\mathbb{R}^n$. A further condition $\mathbf{u}(0) = \mathbf{v}$, for some $\mathbf{v}$ in $\mathbb{R}^n$ is called a **initial condition**.

Theorem 34.3. *Let A be an $n \times n$ matrix and $\mathbf{v} \in \mathbb{R}^n$. The solution of the differential equation $\frac{d\mathbf{u}}{dt} = A\mathbf{u}$ with initial condition $\mathbf{u}(0) = \mathbf{v}$ is $\mathbf{u}(t) = e^{At}\mathbf{v}$.*

Theorem 34.4. *Let A be an $n \times n$-matrix, and let $\mathbf{v} \in \mathbb{R}^n$ be an eigenvector of A with eigenvalue λ, then $e^{At}\mathbf{v} = e^{\lambda t}\mathbf{v}$.*

Theorem 34.5. *Let A be an $n \times n$-matrix and $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be an eigenbasis of A with eigenvalues $\lambda_1, \ldots, \lambda_n$. If $\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$, then the unique solution to the differential equation $\frac{d\mathbf{u}}{dt} = A\mathbf{u}$ with initial condition $\mathbf{u}(0) = \mathbf{v}$ is*

$$e^{At}\mathbf{v} = c_1 e^{\lambda_1 t}\mathbf{v}_1 + \cdots + c_n e^{\lambda_n t}\mathbf{v}_n.$$